

Homework 5

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1 Introduction

In this homework I model a thin elastic plate using the Discrete Elastic Shell (DES) formulation and compare its static deflection under gravity to the Euler–Bernoulli beam prediction. The plate is discretized into two rows of mass points connected by stretching springs and bending hinges. I integrate the equations of motion in time with damping until the motion dies out, interpret the final configuration as a static equilibrium under self-weight, and compare the resulting tip displacement δ_{plate} to the analytical Euler–Bernoulli cantilever deflection δ_{EB} .

2 Discrete shell model

2.1 Geometry and discretization

The plate has length $l = 0.1$ m, width $w = 0.01$ m, and thickness $h = 0.002$ m. I place two rows of nodes in the x – y plane at $z = 0$ with

$$x \in \{-0.0125, 0, 0.0125, 0.025, 0.0375, 0.05, 0.0625, 0.075, 0.0875, 0.1\} \text{ m}, \quad y \in \{0, 0.01\} \text{ m},$$

yielding $n_v = 20$ nodes in total.

The four leftmost nodes are clamped, while all other nodes are free to move. Gravity acts in the $-z$ direction.

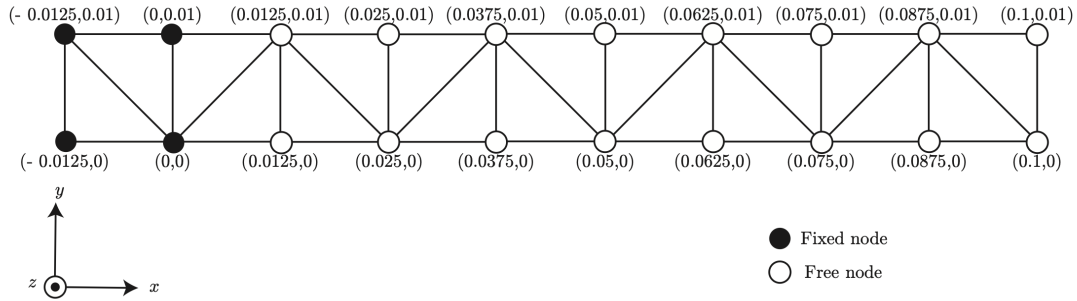


Figure 1: Initial discrete shell configuration in the x – y plane. Black lines denote stretching edges; the four fixed nodes at the left end are highlighted.

2.2 Stretching and bending elements

The in-plane behavior is modeled with stretching springs placed on all edges of a triangulated mesh. Starting from the 2×10 grid of nodes, each four-node cell with bottom-left, bottom-right, top-left, and top-right nodes (bl, br, tl, tr) is split into two triangles with an *alternating* diagonal pattern, as in the HW5 handout: for even cells the diagonal connects (br, tl), while for odd cells the diagonal connects (bl, tr). From this list of triangles I extract all unique undirected edges and attach a stretching spring

to each of them, so that springs lie along all bottom and top horizontals, the vertical connections between the two rows, and the alternating diagonals.

Out-of-plane bending is modeled with discrete hinges located on every *interior* edge of the triangulation, i.e. every edge that is shared by exactly two triangles. Programmatically, for each undirected edge (a, b) I find the incident triangles (a, b, c_0) and (a, b, c_1) ; if an edge has two such triangles, I create a hinge

$$(a, b, c_0, c_1),$$

where (a, b) is the hinge edge and c_0, c_1 are the opposite vertices. In the present 2×10 strip this construction produces hinges along all alternating diagonal edges and along the interior vertical edges between the two rows, while boundary edges (used by only a single triangle) do not carry hinges. For each hinge I compute the signed dihedral angle

$$\theta = \theta(x_0, x_1, x_2, x_3)$$

using the DES hinge routine, and I use the flat configuration to define the natural angle θ_{bar} .

2.3 Material parameters and element stiffness

The plate has Young's modulus $Y = 10^7$ Pa and density $\rho = 1,000$ kg/m³. The stretching stiffness on edge k with reference length l_k is

$$k_{s,k} = \frac{\sqrt{3}}{2} Y h l_k^2,$$

and the bending stiffness of each hinge is

$$k_b = \frac{2}{\sqrt{3}} \frac{Y h^3}{12},$$

which are the standard DES formulas obtained by matching the continuum membrane and bending energies of a triangular strip.

The total plate mass $M = \rho w h l$ is lumped equally over the n_v nodes, so that each node has mass $m_i = M/n_v$ and the mass matrix is diagonal,

$$\mathbf{M} = \text{diag}(m_1, \dots, m_{n_v}).$$

3 Numerical time integration

3.1 Equations of motion and Newton solve

Let $q^n \in R^{3n_v}$ denote the stacked nodal positions at time step n , and u^n the velocities. I use the semi-implicit scheme from the HW template. At each step I solve for q^{n+1} by enforcing the discrete equations of motion on the free DOFs:

$$\mathbf{f}(q^{n+1}) = \frac{\mathbf{m}}{\Delta t} \left(\frac{q^{n+1} - q^n}{\Delta t} - u^n \right) - (\mathbf{F}_s + \mathbf{F}_b + \mathbf{F}_g + \mathbf{F}_v) = 0,$$

where $\mathbf{F}_s$ and $\mathbf{F}_b$ are the stretching and bending forces, $\mathbf{F}_g$ is the gravitational force, and

$$\mathbf{F}_v = -\nu \frac{q^{n+1} - q^n}{\Delta t}$$

is a viscous damping term with $\nu = 0.01$. The Jacobian $\partial \mathbf{f} / \partial q$ is assembled from the element Hessians, and the system is solved via Newton–Raphson, restricting to the free DOFs by eliminating the rows/columns corresponding to fixed nodes.

I use a time step $\Delta t = 10^{-3}$ s and integrate up to $T = 5$ s. At each step the Newton iteration is terminated when the residual on the free DOFs falls below a prescribed tolerance.

After solving for q^{n+1} , the velocity is updated as

$$u^{n+1} = \frac{q^{n+1} - q^n}{\Delta t}.$$

3.2 Tip displacement

The quantity of interest is the vertical displacement of the rightmost bottom node at $(x, y) = (0.1, 0)$, which I denote by $\delta_{\text{plate}}(t)$:

$$\delta_{\text{plate}}(t^n) = z_{\text{tip}}(t^n) - z_{\text{tip}}(0),$$

where $t^n = n\Delta t$. The time history of $\delta_{\text{plate}}(t)$ is shown in Fig. 2. Due to numerical damping, the response decays monotonically and approaches a steady value by about $t \approx 2$ s.

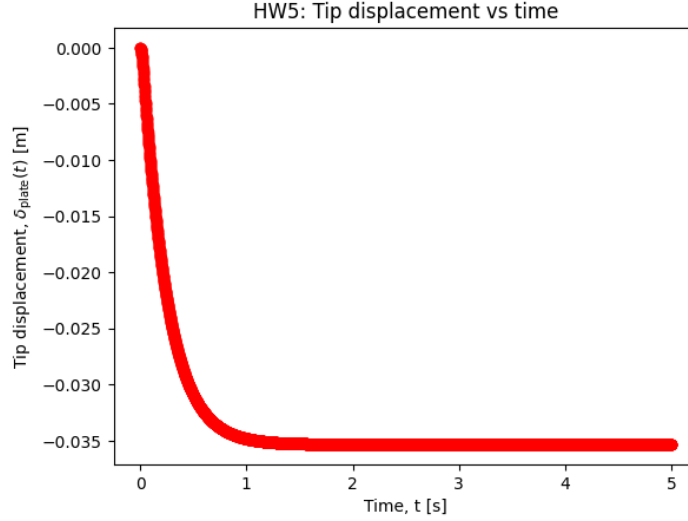


Figure 2: Tip displacement $\delta_{\text{plate}}(t)$ for the discrete shell plate under gravity. The displacement decreases monotonically and levels off to a steady value as the dynamics are damped out.

A few snapshots of the plate configuration are shown in Fig. 3. The plate bends smoothly downward while remaining approximately planar in the x - z plane, consistent with the cantilever beam interpretation.

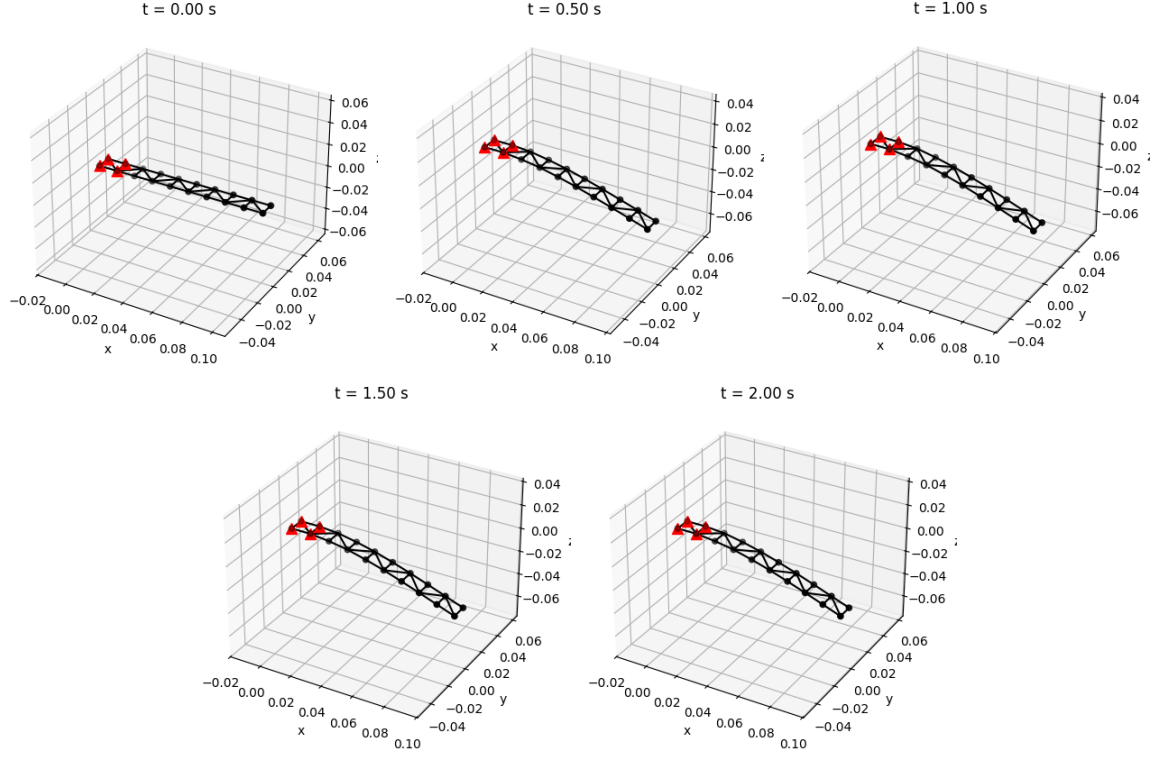


Figure 3: Snapshots of the plate configuration at $t = 0, 0.5, 1.0, 1.5, 2.0$ s. The four fixed root nodes are indicated in red. Under gravity the plate bends downward in a beam-like shape.

To obtain a robust estimate of the static deflection, I average the last portion of the time series (here the final 200 time steps):

$$\bar{\delta}_{\text{plate}} = \frac{1}{200} \sum_{n=N-199}^N \delta_{\text{plate}}(t^n) \approx -3.53 \times 10^{-2} \text{ m},$$

where N is the index of the final time step.

4 Euler–Bernoulli beam comparison

4.1 Analytical prediction

I treat the plate as a uniform Euler–Bernoulli cantilever beam of length l , width w , thickness h , and density ρ . The cross-sectional area and second moment of area are

$$A = wh, \quad I = \frac{wh^3}{12}.$$

Under gravity the beam experiences a uniform distributed load

$$q = \rho Ag.$$

For a cantilever fixed at $x = 0$ and loaded by a uniform load q along $0 \leq x \leq l$, the tip deflection at $x = l$ is

$$\delta_{\text{EB}} = -\frac{ql^4}{8YI},$$

where the negative sign indicates downward deflection in our coordinate system.

Substituting the numerical values used in the simulation gives

$$\delta_{\text{EB}} \approx -3.675 \times 10^{-2} \text{ m}.$$

4.2 Quantitative comparison

Figure 4 overlays the DES tip displacement with the Euler–Bernoulli prediction. The solid curve shows $\delta_{\text{plate}}(t)$, the dashed horizontal line marks the steady simulation value $\bar{\delta}_{\text{plate}}$, and the dotted line marks δ_{EB} .

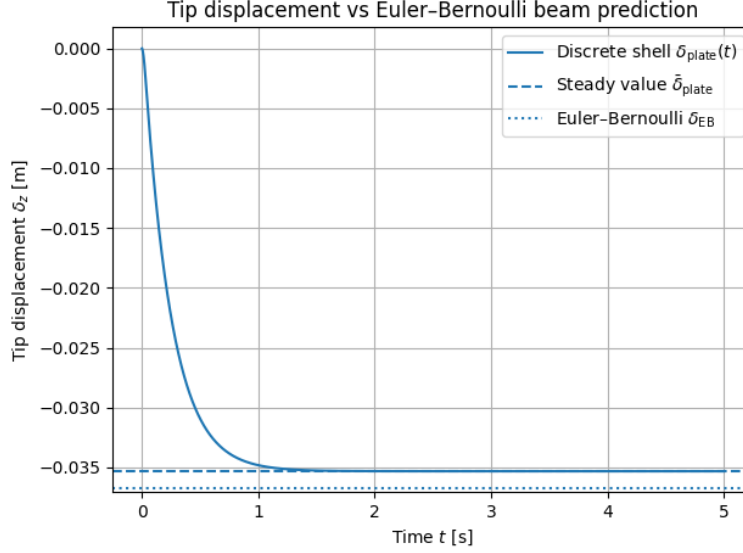


Figure 4: Tip displacement of the discrete shell plate (solid line) compared with the Euler–Bernoulli cantilever prediction (dotted line). The dashed line indicates the steady value $\bar{\delta}_{\text{plate}}$ obtained by averaging the last part of the simulation.

Numerically, the steady DES displacement and analytical prediction are

$$\bar{\delta}_{\text{plate}} \approx -3.53 \times 10^{-2} \text{ m}, \quad \delta_{\text{EB}} \approx -3.68 \times 10^{-2} \text{ m}.$$

The relative difference is

$$\frac{\bar{\delta}_{\text{plate}} - \delta_{\text{EB}}}{\delta_{\text{EB}}} \approx -3.87\%,$$

so the DES model is about 4% *stiffer* than the ideal Euler–Bernoulli beam (its magnitude of deflection is slightly smaller).

This level of agreement is reasonable given that:

- the plate is represented by only two nodes across its width and a relatively coarse discretization along the length;
- the discrete hinge stiffness is matched only approximately to the continuum bending stiffness;
- clamping four nodes at the root introduces a slightly more constrained boundary condition than the ideal continuum cantilever.

5 Conclusions

Using the discrete elastic shell formulation with stretching and bending energies, I simulated a thin plate under its own weight and treated it as a cantilever beam. The plate bends smoothly into a beam-like shape, and the tip displacement converges to a steady value of approximately $\bar{\delta}_{\text{plate}} \approx -3.53 \text{ cm}$. The Euler–Bernoulli prediction for the corresponding cantilever gives $\delta_{\text{EB}} \approx -3.68 \text{ cm}$, resulting in a relative difference of about 4%. This shows that even a coarse DES model with only two nodes across the width can capture the correct bending compliance of a thin beam to within a few percent, while also providing access to the full 3D geometry and dynamics of the structure.